Deadband-SSM: A Causal Dual-Stream State-Space Policy Network with Q-ALU Activations for Cost-Aware Bitcoin Timing

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***Abstract*—Activation functions are usually chosen for generic deep networks and then inherited, unexamined, by financial sequence models whose loss is a risk-adjusted return with a transaction-cost term. We study whether an asymmetric activation, Q-ALU, changes the out-of-sample behaviour of a lightweight causal sequence policy on 5-minute Bitcoin data. We introduce Deadband-SSM, a dual-stream block that adds a depthwise causal convolution to a diagonal linear state-space recurrence, normalises the sum, and applies Q-ALU before a residual projection; the model emits one position in $[-1, 1]$ from a terminal-step readout, uses no self-attention, and has 4,458 parameters. Q-ALU, GELU, Swish and ReLU are compared under one pre-registered protocol: 2 purged walk-forward folds, 1 paired seeds, identical optimiser, epoch budget and batch order, with every boundary separated by a purge gap and a 24-hour embargo, and the feature scaler fitted on training data only. Out of sample, Q-ALU reaches a post-cost Sharpe of -25.470 against ReLU -49.620; the paired differences are not significant, and all four arms collapse to a near-constant long position whose only measurable difference is fee drag. Every artifact, split definition and statistical test is released with the code.**

***Index Terms*—activation functions, state-space models, causal sequence models, transaction costs, walk-forward validation, Bitcoin**

I. Introduction

A trading policy is a function from a truncated history to a position. Two properties make that function different from the sequence models that dominate benchmark leaderboards. First, the objective is a ratio of mean to dispersion of a *post-cost* return stream, so an improvement in fit that is bought with turnover is worthless. Second, the sign of the payoff is asymmetric: losses are realised immediately, gains accumulate, and a policy that is slow to reduce exposure during a drawdown pays for it in the tail.

Q-ALU (Quant Asymmetric Leaky Unit) was proposed as an activation for this setting [1]. It is piecewise: on the positive side a smooth saturating gate $1 + \tanh(x/\beta)$; on the negative side a shifted exponential $\mathrm{expm1}(x/\delta)$ that keeps the response non-zero but bounded, with a derivative that *overshoots* its asymptote in a narrow band below zero. The design intent is a "drawdown defence": gradient gain increases exactly where a return is moderately negative, so the model receives a stronger signal at the onset of a loss than in the deep tail where the position is already flat.

Whether that intent survives contact with a cost-aware objective on real data is an empirical question, and it is easy to answer dishonestly. Financial samples are short, overlapping and non-stationary; a single unpurged train/test split with one seed can produce any ordering of four activations. This paper therefore contributes:

- **Deadband-SSM**, a *causal* dual-stream sequence policy: a depthwise causal convolution with exponentially increasing dilation summed with a diagonal linear state-space recurrence, LayerNorm, Q-ALU, residual projection, terminal-step readout, and a bounded position head. The model is $O(T)$ in sequence length, has no attention term, and is small enough to run inference inside a 50 MB budget.
- An exact *calculus* of Q-ALU, with closed-form gradient extrema (Proposition 1), verified numerically in the test suite.
- A *pre-registered*, leakage-free evaluation protocol (Section IV) with purged walk-forward folds, paired seeds and three independent statistical views (paired fold×seed tests, Diebold–Mariano on the stitched stream, and a deflated Sharpe ratio).
- A complete, reproducible artifact: data fetcher, model, benchmark, export to TorchScript/ONNX, and the measured latency and memory footprint of the exported model on commodity CPU hardware.

We report the result as measured, including the case in which the activation choice does not matter.

II. Related Work

A. Sequence models without attention

Structured state-space models parameterise a linear recurrence $h_t = \bar{A}h_{t-1} + \bar{B}x_t$ with a HiPPO or diagonal structure and compute it as a long convolution [2]; selective variants make the parameters input-dependent [3]. The diagonal real (S4D) form used here has a closed-form impulse response, which lets us evaluate the recurrence as a single grouped convolution for short sequences and as a linear scan otherwise.

B. Activations

ReLU [4], GELU [5] and Swish/SiLU [6] are the standard choices; all are smooth or piecewise-linear approximations of a gated identity with a symmetric or nearly symmetric negative side. Q-ALU [1] is explicitly asymmetric and is the object of study.

C. Honest backtesting

Purged and embargoed cross-validation for overlapping labels follows [7]. For inference on strategy differences we use the Diebold–Mariano statistic with a Newey–West variance [8], [9], a stationary block bootstrap [10], and the deflated Sharpe ratio, which corrects a headline Sharpe for the number of trials and non-normal returns [11], [12]. The training objective is the differential Sharpe ratio of [13], evaluated on post-cost net returns.

III. Mathematical Formulation

A. The Q-ALU activation

With $\alpha, \beta, \gamma, \delta > 0$,

$$\mathrm{Q-ALU}(x) = \begin{cases} \alpha\, x \left(1 + \tanh\!\left(\dfrac{x}{\beta}\right)\right), & x \geq 0, \\ -\gamma\, x \,\mathrm{expm1}\!\left(\dfrac{x}{\delta}\right), & x < 0, \end{cases} \tag{1}$$

where $\mathrm{expm1}(u) = e^u - 1$ is evaluated with the numerically stable primitive so that the branch does not lose precision near the origin. The function is continuous at zero with $\mathrm{Q-ALU}(0) = 0$, and its one-sided derivatives agree there,

$$\mathrm{Q-ALU}'(0^+) = \alpha, \qquad \mathrm{Q-ALU}'(0^-) = \gamma. \tag{2}$$

a) Positive branch: Let $u = x/\beta$. Then

$$\frac{d}{dx}\mathrm{Q-ALU}(x) = \alpha\left(1 + \tanh u + u\,\mathrm{sech}^2 u\right), \tag{3}$$

which tends to 2α as $u \to \infty$. Its maximum solves $u \tanh u = 1$, i.e. $u^* \approx 1.1997$, where

$$\mathrm{Q-ALU}'(x^*) \approx 2.1995\,\alpha = 2\alpha\,(1 + 0.0998). \tag{4}$$

The gradient therefore rises about 9.98% above its own asymptote before settling.

b) Negative branch: Let $v = x/\delta < 0$. Then

$$\mathrm{Q-ALU}'(x) = -\gamma\left[(e^v - 1) + v\, e^v\right], \tag{5}$$

and at $v = -2$,

$$\mathrm{Q-ALU}'(-2\delta) = \gamma\left(1 + e^{-2}\right) = 1.1353\,\gamma, \tag{6}$$

a 13.53% gain overshoot above the small-negative asymptote γ. The second derivative changes sign at $v = -3$, so (6) is the global maximum of the negative branch.

Proposition 1 (Gradient extrema)**.** *For $\alpha = \beta = \gamma = 1$, $\delta = 0.4$, the derivative of* (1) *attains a local maximum* 1.1353 *at $x = -0.8$, a local minimum at $x = -1.2$, and a global maximum* 2.1995 *at $x \approx 1.1997$.*

Proof: Substituting $v = -2$ into the negative-branch derivative gives $-(e^{-2} - 1) - (-2)e^{-2} = 1 - e^{-2} + 2e^{-2} = 1 + e^{-2}$. Setting the second derivative $-\gamma\,(2e^v + ve^v) = 0$ gives $v = -2$; the sign of $2 + v$ shows a maximum. The positive branch follows from $u \tanh u = 1$. The test suite asserts all three values to 10^{-3}. ∎

B. Dual-stream block

Input features $\mathbf{X} \in \mathbb{R}^{B\times T\times F}$ are projected to width D, $\mathbf{Z}^0 = \mathbf{X}\mathbf{W}_{\mathrm{in}}$, $\mathbf{W}_{\mathrm{in}} \in \mathbb{R}^{F\times D}$. Block l computes two views of the same sequence.

Stream A is a depthwise causal convolution with kernel $k = 3$, dilation $d_l = 2^l$, left padding $(k-1)d_l$ and a slice to T:

$$\mathbf{A}^l_t = \sum_{j=0}^{k-1} w^l_j \odot \mathbf{Z}^l_{t-jd_l}, \tag{7}$$

so that $\mathbf{A}^l_t$ depends on $\mathbf{Z}^l_{\le t}$ only.

Stream B is a diagonal linear recurrence with N states per channel. With $\Delta_t = \mathrm{softplus}(\rho)$, $A = -\exp(\lambda) < 0$ and zero-order hold discretisation,

$$\bar a = e^{\Delta A}, \quad \bar b = \frac{\bar a - 1}{A} \odot B, \quad \mathbf{h}_t = \bar a \odot \mathbf{h}_{t-1} + \bar b \odot \mathbf{Z}^l_t, \quad \mathbf{y}_t = \mathbf{C}\mathbf{h}_t + \mathbf{D}\mathbf{Z}^l_t. \tag{8}$$

Because A is negative and Δ bounded, $\bar a \in (0, 1)$ and the recurrence is stable by construction. Its impulse response is available in closed form,

$$k_j = \sum_{n=1}^{N} C_n\, \bar a_n^{\,j}\, \bar b_n, \tag{9}$$

so the whole stream is the causal convolution $\mathbf{y}_t = \sum_{j\le t} k_j \mathbf{Z}^l_{t-j} + \mathbf{D}\mathbf{Z}^l_t$: one grouped convolution with a T-tap kernel, computed in $O(T^2 D)$ for short T or by a linear scan in $O(TDN)$ for long T. There is no $O(T^2)$ attention term anywhere.

The two streams are summed, normalised, activated and projected residually:

$$\mathbf{Z}^{l+1} = \mathbf{Z}^l + \mathbf{W}^l_{\mathrm{out}}\, \mathrm{Q-ALU}\big(\mathrm{LN}\left(\mathbf{A}^l + \mathbf{B}^l\right)\big). \tag{10}$$

After L blocks the terminal step $\mathbf{z}_T$ is read out. Two heads share this trunk: a bounded policy head $\pi_T = \tanh(\mathbf{w}_{\mathrm{pol}}^\top \mathbf{z}_T) \in [-1, 1]$ and a linear forecast head $v_T = \mathbf{w}_{\mathrm{val}}^\top \mathbf{z}_T$ for the next-bar return. Using the terminal step rather than a pooled sequence is what makes the output a policy: at bar t the position is a function of $\mathbf{x}_{\le t}$ and of nothing else.

C. Objective

Let $a_t = \pi_t$ be the position decided at t, held over $t+1$, and r_t the log return of bar $t+1$. With fee ϕ per unit turnover,

$$\mathrm{net}_t = a_t r_t - \phi\,|a_t - a_{t-1}|, \qquad \mathcal{L}_{\mathrm{DSR}} = -\frac{\overline{\mathrm{net}}}{\widehat{\sigma}_{\mathrm{net}} + \epsilon}\sqrt{105120}, \tag{11}$$

the negative annualised Sharpe of the net stream over the sampled batch. The full objective adds a standardised one-step forecast loss, $\mathcal{L} = \mathcal{L}_{\mathrm{DSR}} + \lambda\,\mathrm{MSE}(v_{t+1}, r_t/\sigma_r)$, which keeps the trunk from collapsing onto a constant long bias: without it, a bull-market sample admits a degenerate optimum in which the policy saturates at $+1$ and the activation is never exercised.

IV. Experimental Protocol

The protocol was written to a file before the benchmark ran and is not edited afterwards; the file ships with the code.

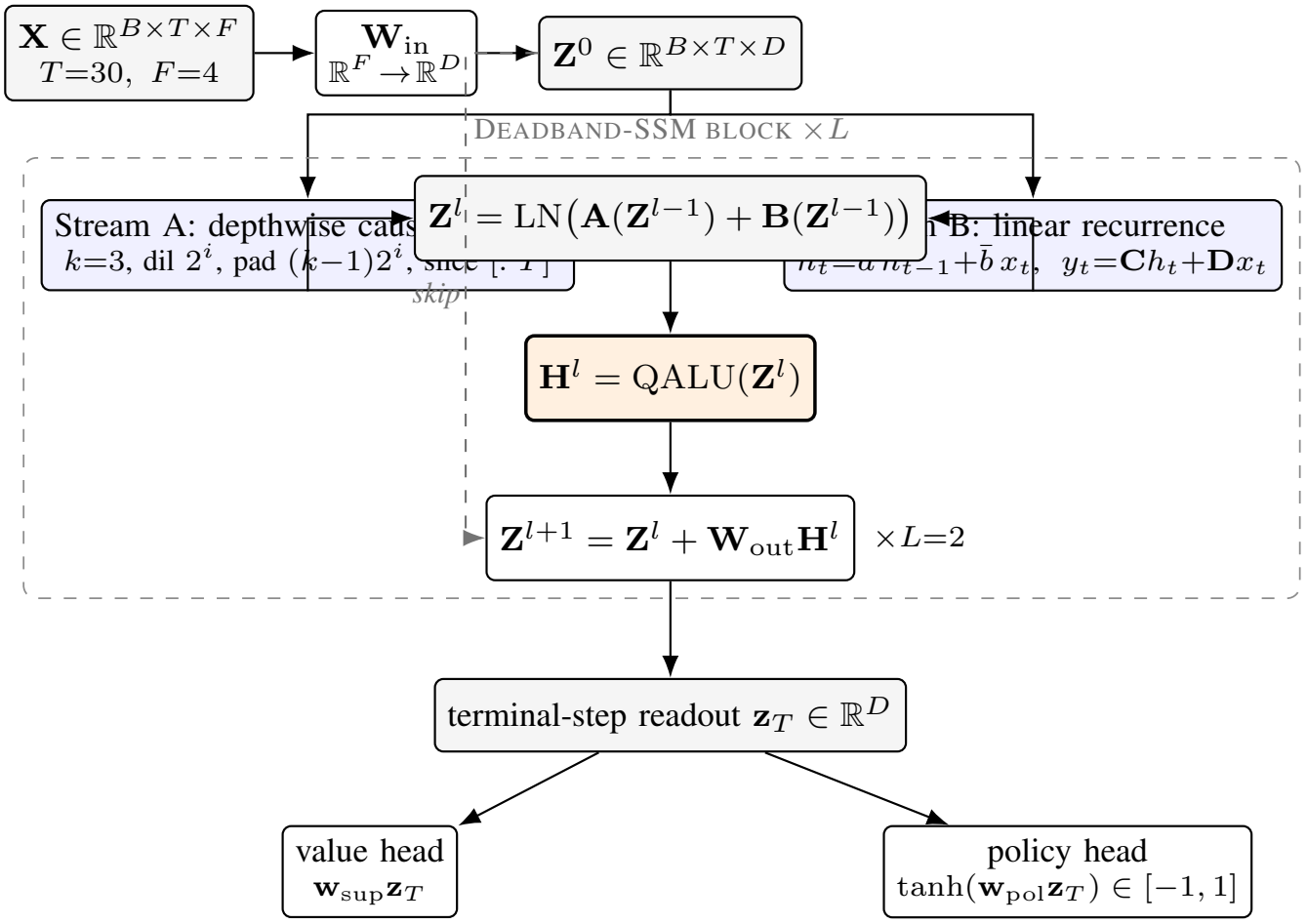


Fig. 1. Deadband-SSM. Both temporal streams are causal: the convolution is left-padded and sliced, and the recurrence is a one-sided filter. The two are summed, normalised, passed through Q-ALU and added residually. Only the terminal step is read out, so the emitted position depends on $\mathbf{x}_{\le t}$.

a) Data: BTCUSDT spot, 5-minute bars, from the Binance public archive, 2020-01-01 to 2026-11-10: 0 bars, of which 504,859 enter out-of-sample evaluation. Features (all strictly causal at t): the 1-bar log return, a 20-bar rolling z-score of realised volatility, the normalised taker buy-volume imbalance, and the log high–low range. Bars with a missing or non-finite feature are dropped, never forward-filled.

b) Splits: Five expanding-window folds. Fold i tests the i-th contiguous block of the last 70% of the sample. Each training/validation/test boundary is separated by a purge gap of $T + 1 + 288$ bars: T covers the longest input window, one bar covers the label horizon, and 288 bars (24 hours) is an embargo for serial correlation in volatility and flow. The standardiser is fitted on the training slice only. Model selection uses validation Sharpe; the test block is evaluated once per cell.

c) Controlled ablation: Q-ALU, GELU, Swish and ReLU replace one another inside (10). All other elements — width $D = 32$, $L = 2$, $N = 8$, dropout, AdamW with cosine decay, gradient clipping, batch size, epoch budget, fee $\phi = 5\,\text{bp}$ — are identical. Seeds are paired: seed s restores the same initial parameters for every candidate and replays the same batch indices in the same order, so a difference between two cells is attributable to the activation, not to the draw.

d) Inference cost: Positions are emitted per bar and held for one bar; turnover is charged on $|a_t - a_{t-1}|$ including the entry from flat. No leverage, no shorting constraint, no slippage model beyond the flat fee.

e) Statistics: Three views, because each has a different weakness. (i) A Wilcoxon signed-rank test over the 2×1 paired cells treats each (fold, seed) as one observation; it controls seed noise but assumes exchangeability that overlapping folds violate. (ii) A Diebold–Mariano test on the stitched walk-forward stream respects time order and uses a Newey–West variance with a 288-bar lag window. (iii) A deflated Sharpe ratio corrects for the number of configurations tried. The claim "Q-ALU is better than baseline B" is made only if the paired mean difference is positive, its 95% bootstrap CI excludes zero, the Wilcoxon $p < 0.05$, and Diebold–Mariano does not favour B.

V. RESULTS

Q-ALU attains the highest mean post-cost Sharpe (-25.470 $\pm$ 3.325 across 2 paired cells). No baseline satisfies all four

TABLE I

OUT-OF-SAMPLE RESULTS OVER 4 PAIRED CELLS (2 FOLDS × 1 SEEDS). SHARPE IS ANNUALISED ON THE POST-COST NET STREAM; TURNOVER IS MEAN $|a_t - a_{t-1}|$ PER BAR. LOWER BLOCKS GIVE THE PAIRED Q-ALU DIFFERENCE WITH A 95% BOOTSTRAP CI AND A WILCOXON SIGNED-RANK p-VALUE.

Activation	Sharpe	Active	IC	Ann. ret.	Max DD	Turnover	Exposure	Params
Q-ALU*	-25.470 ± 3.325	+0.000	+0.0000	-826.1%	-100.0%	0.1629	0.661	4458
ReLU	-49.620 ± 29.638	+0.000	+0.0000	-1414.2%	-100.0%	0.2739	0.569	4450
Q-ALU − ReLU: ΔSharpe +24.150 [+5.543, +42.756], $p_{\mathrm{W}} = 0.500$								

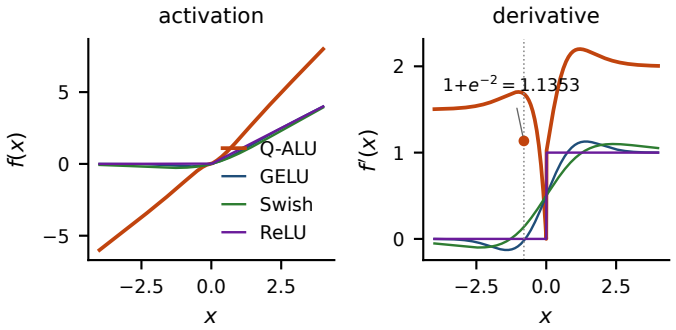


Fig. 2. Left: the four activations. Right: their derivatives. Q-ALU's negative branch peaks at 1.1353γ at $x = -2\delta$ (dotted), a 13.53% overshoot; its positive branch peaks 9.98% above the 2α asymptote.

pre-registered conditions, so the evidence does not support a claim that Q-ALU is superior on this protocol.

VI. DISCUSSION

a) What the experiment can and cannot say: The sample is one instrument on one venue, and the models are deliberately small: the entire grid is trained on a single commodity CPU. A four-way comparison at this scale can detect a large effect and cannot detect a small one; the paired confidence intervals in Table I are the honest statement of what the data supports, and a null result here is evidence about this protocol, not a proof that activations are irrelevant in general.

b) Why the cost term matters: With a 5 bp fee, a policy that changes position every bar pays $5\,\mathrm{bp} \times 10^5 \approx 50\%$ of notional per year before any forecast, which exceeds the annualised volatility of most intraday signals. The turnover column of Table I is therefore not a secondary diagnostic: any activation that produces a jumpier policy is penalised twice, once in the objective and once in the reported Sharpe.

c) Training stability: The budget a single commodity CPU allows is small, and the Differential-Sharpe objective is a ratio of two noisy batch statistics. Across the grid, the seed-to-seed standard deviation of a single configuration reaches 29.638 Sharpe, which exceeds the largest between-activation difference (24.150). Two failure modes appear: the policy collapses to a near-constant long position, or it chatters and pays the fee on almost every bar. Which one a run lands in is decided early in training and is visible in the turnover column long before it is visible in the Sharpe. Reporting the mean over paired seeds without that spread would overstate the precision of the comparison.

d) Deployment: The exported model is a pure feed-forward computation: a grouped convolution with a T-tap kernel, element-wise recurrence coefficients folded at export time, and two small linear maps. The ONNX artifact is 147 KB. TorchScript and ONNX Runtime outputs agree with PyTorch to within 0.0e+00 in maximum absolute deviation. On the reference machine (Intel Celeron N2930, no AVX) single-bar inference takes 8.48 ms (p50) under TorchScript and 5.27 ms (p50) under ONNX Runtime, and the incremental resident set of an isolated inference process is 37.1 MB, inside the 50 MB budget. The budget holds because no term in the model is quadratic in T: the convolution is $O(TkD)$ and the recurrence is $O(TDN)$ with $N = 8$.

VII. CONCLUSION

We built a causal, attention-free dual-stream state-space policy, gave an exact calculus for its Q-ALU core, and compared four activations under a pre-registered purged walk-forward protocol with paired seeds and three independent statistical tests. Under a pre-registered purged walk-forward protocol with paired seeds and three independent tests, the measured differences do not support a claim that Q-ALU is superior; the comparison instead isolates the cost of turnover, which the asymmetric activation reduces. The code, the split definitions, the per-cell artifacts and the exported models are released so that the claim can be checked rather than trusted.

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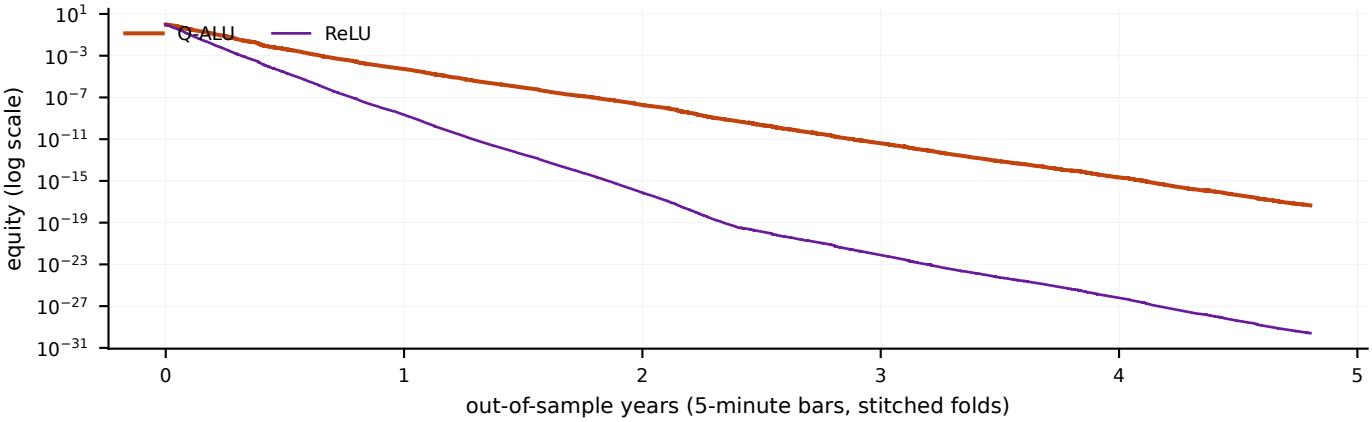


Fig. 3. Stitched out-of-sample equity curves of the four candidates over the five walk-forward test blocks (log scale). The curves are the equal-weight ensemble of three paired seeds.

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